MACS 30111 · Python vs NumPy vs pandas

Name: _____ Nov 12, 2026

Complete the full worksheet (front and back) for 5 points. Incomplete worksheets earn 3 points.

My column: ☐ base Python ☐ NumPy ☐ pandas

1. **On your own:** do only your column in Part 1 (front) and Part 3 (back).
2. **In your group of three:** explain your column to the others, and fill in their columns on your sheet.
3. **Together:** do Part 2. The base Python person leads the group's check.
4. **Compare** with a neighboring group and fix anything you disagree on.

Part 1 · What can each one do?

```
x   = [8, 10, 9, 7]              # base Python list
arr = np.array([8, 10, 9, 7])    # NumPy array
s   = pd.Series([8, 10, 9, 7], index=["Dwight", "Angela", "Pam", "Jim"])   # pandas Series
```

For each row, put a ✓ in each column that can do the job **in one step**: a single operator, function or method, with **no loop you write yourself**. Base Python means lists, dictionaries and built-ins, with no `import`. **If you check it, write the code that does it.** Leave the box blank if it can't be done in one step.

	I want to...	**Base Python** list / dict	**NumPy** array	**pandas** Series / DataFrame
ex	**Example:** Count how many values there are get `4`	✓ `len(x)`	✓ `len(arr)`	✓ `len(s)`
1	Hold text and numbers together in one object e.g., `"Pam"` and `9`			✔
2	Get a value by its position e.g., the first score, `8`	✔	✔	✔
3	Get a value by a name (label) e.g., Pam's score, using `"Pam"`	✔		✔
4	Change one value in place e.g., change the first score to `99`	✔	✔	✔
5	Add 1 to every value in one step get `9, 11, 10, 8` with no loop		✔	✔
6	Keep only the values greater than 8 with a Boolean mask keep `10, 9`		✔	✔
7	Sort the values get `7, 8, 9, 10`	✔	✔	✔
8	Mark which values are missing (NaN) True where a value is missing		✔	✔
9	Count how many times each value appears e.g., grades `A, B, A` → `A: 2, B: 1`		✔ unique	✔ value_counts
10	Store a table (rows × columns) and pull out one column e.g., SE1 and SE2 scores; get just SE2		✔	✔
11	Group rows and summarize each group e.g., average `tree_dbh` for each borough			✔
12	Read a CSV with text and number columns into a table e.g., the tree census			✔
13	Combine two tables by matching a shared column e.g., add lifespans by species			✔ merge

Turn over →

Part 2 · Verb check

Do this together; the base Python person leads. Which library is this verb a job for? Put a ✓ under NumPy, pandas or both. Then write the function, method or operator that does it (e.g., average → `.mean()`).

	Verb	NumPy	pandas	How to write it
1	group		✔	df.groupby
2	sort	✔	✔	np.sort, sort_values
3	merge two tables		✔	
4	broadcast	✔		(sort of also with Pandas!)
5	pivot (long → wide)		✔	
6	count each value		✔	value_counts (could make numpy arg)
7	fill in missing values		✔	fillna
8	drop missing rows		✔	dropna
9	matrix-multiply	✔		
10	random numbers	✔		
11	filter with a mask	✔	✔	arr[arr>13], df[df[“ upvotes ”] <42]
12	label rows with names		✔	
13	read a mixed CSV		✔	pd.read_csv
14	transpose	✔	✔	.T

Part 3 · Same task, three ways

Write one line of code for each task in each column. Start from the line at the top of the column and assume `import numpy as np` and `import pandas as pd`. Your code should produce the result shown. The first row is done for you.

Reminder: in base Python, math and comparisons don't apply to every item of a list at once. Use a list comprehension like `[s * 2 for s in scores]` or a loop. In NumPy and pandas, operators like `+` and `>=` work on every value at once.

	Base Python `scores = [8, 10, 9, 7]`	**NumPy** `arr = np.array([8, 10, 9, 7])`	**pandas** `s = pd.Series([8, 10, 9, 7], index=["Dwight", "Angela", "Pam", "Jim"])`
Example: Get Pam's score Result: `9`	`scores[2]`	`arr[2]`	`s["Pam"]`
Add 1 to every score Result: `[9, 11, 10, 8]`	[s+1 for s in scores]	arr + 1	s + 1
Keep only the scores of 9 or more Result: `[10, 9]`	[s+1 for s in scores if s >8]	arr[arr >8]	s[s>8]
Find the average score Result: `8.5`	sum(scores)/len (scores)	arr.mean()	s.mean()

Which column took the most work, and why? _____

_____